

ViSenze: Provides AI- based solutions and powerful data analysis to improve ecommerce revenue

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ABOUT VISENZE

The ViSenze logo features the word "VISENZE" in a blue, sans-serif font. A small red dot is positioned above the letter "i".

About ViSenze

With Google Cloud, ViSenze has created a data platform that can ingest and process 500 million records and store data for a year, while giving technical team members

ViSenze is the world's smartest product discovery platform. Retailers like ASOS, H&M, Rakuten, Zalora, Meesho, DFS, and EyeBuyDirect trust ViSenze to power their product discovery. ViSenze processes over a billion queries a month from retailers, supporting them in-store and on ecommerce sites. Use cases range from enhanced search, product tagging, smart recommendations and merchandising analytics. ViSenze's automated AI platform empowers retail merchants to increase revenue by

the ability to generate
detailed, insightful

delivering exceptional experiences that make it
easier for shoppers to discover products.

- Gartner Hype Cycle for Retail Technologies 2021
- CB Insights Retail Technology 100
- Top AI Companies 2022
- Leading Product Recommendation Vendor 2022

Google Cloud results

- Reduced data infrastructure and management costs by 50%
- Scaled analytics data warehouse without incurring downtime
- Enabled non-technical team members to create detailed reports

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data
analy

Industries: Technology

Location: Singapore

(<https://marketingplatform.google.com/about/d>

BigQuery (<https://cloud.google.com/bigquery/>)

, [Cloud SQL](https://cloud.google.com/sql/) (<https://cloud.google.com/sql/>),

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

(<https://cloud.google.com/compute/>) Cloud

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

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Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

AI, deep
learning, and
image
recognition is

transforming
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These
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rather than text
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Visual
commerce
provider

ViSenze

(<https://www.visenze.com>)

is helping some
of the world's

leading retailers
improve
conversion
rates through
image-based
search.

The business's products also enable media companies to use the platform to turn images and videos into engagement opportunities—driving new and incremental revenues.

Created through NExT—a research center established by the National University of Singapore and Tsinghua University of China—ViSenze now operates in the United States, United Kingdom, India, China, and Singapore. The business is backed by Japan-based internet and ecommerce company Rakuten and cross-border investment specialist WI Harper Group.

Growth in SMB and mobiles

Renjie Yao, Data Platform Lead at ViSenze, sees opportunities for growth in the small-to-medium business sector, where companies do not have the resources to build similar technologies, and with mobile device OEMs to integrate ViSenze natively on smartphones. Phone owners can activate a “shopping lens” on camera and gallery apps to capture an image of a product. They then receive matching results from more than 800 partner merchants and retailers and can then click through to product pages on partner apps or mobile websites. Alternatively, they may use a photo to

compare products sold on different sites or shop matching styles.

The ViSenze API analyzes the contents of a selected or clicked image and sends the information back to the organization's visual commerce platform. The platform feeds back similar results based on that information.

The ViSenze offering also extends to image analysis for the tagging of product attributes—such as a white turtleneck cardigan with full sleeves—to provide an improved search experience.

Data vital to ViSenze

Capturing and analyzing large volumes of data is integral to ViSenze. “We have to understand how consumers interact with our customers’ ecommerce websites and apps,” says Yao. “For example, we need to know who has looked at a particular pair of jeans on a website and whether that visit led to a conversion. We can then tell that customer whether they need to make more stock available.”

ViSenze also relies on data to provide high-quality training for its image recognition models and its domain-specific models for online retail.

Protect customer data

Data is vital to ViSenze—but customer privacy is most important. “All the data we collect is transparent to our customers, meaning they can decide what they do not want us to collect. In

addition, all personal data processing complies with privacy protection regulations in each region, such as the General Data Protection Regulation in Europe.”

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—Renjie Yao, Data
Platform Lead, ViSenze

A quick move to the cloud

ViSenze started operations using servers, storage, networking, and associated systems in an on-

premises data center operated by NExT. However, to support rapid growth, the business decided to move its workloads to the cloud. ViSenze opted for a multi-cloud architecture, using in part a Google Cloud (<https://cloud.google.com/>) data infrastructure.

“Our research found Google Cloud provided a complete, integrated ecosystem rather than a disparate collection of tools and components, and so was ideal for our needs,” says Yao. “We could connect different components with the click of a mouse.” Further, the business found it could easily configure rules and pipelines to route data logs to relevant Google Cloud services.

The review found Google Cloud’s extensive managed services would also remove administration and maintenance tasks from ViSenze’s in-house technology team—freeing team members to focus on more valuable tasks.

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In addition, Google Cloud provided the security features—including custom hardware running hardened operating systems and file systems and encryption of data at rest and in transit—needed to protect sensitive information. Finally, the location of Google Cloud regions in several countries would enable the business to meet regulatory and data sovereignty requirements.

A three-month implementation

ViSenze opted to move to Google Cloud in mid-2017 and completed a three-month implementation using internal resources. “The process was very smooth and intuitive, and we had no problems building our entire data platform within Google Cloud,” says Yao.

The business now uses an architecture comprising Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>) to manage and orchestrate Docker containers running in Google Cloud Platform; BigQuery

(<https://cloud.google.com/bigquery/>) to provide an analytics data warehouse, with Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

providing customizable visualization and reports;

Stackdriver (<https://cloud.google.com/stackdriver/>) to monitor and manage virtual machine instances and services inside Google Cloud; Cloud SQL (<https://cloud.google.com/sql/>) to manage its relational databases for real-time analytics; Compute Engine (<https://cloud.google.com/compute/>) to provide compute resources; Cloud Storage (<https://cloud.google.com/storage/>) to store files and objects; Cloud Pub/Sub (<https://cloud.google.com/pubsub/>) to provide real-time messaging between applications; and Cloud Functions (<https://cloud.google.com/functions/>) to build event-driven applications.

After collecting the request logs of users in virtual machine instances and Docker containers, ViSenze distributes them in three directions. “We export raw logs into Cloud Pub/Sub for indexing inside an Elasticsearch search engine, and to a BigQuery data warehouse for further analytics,” explains Yao. “We also use Cloud Functions-created applications to obtain the logs from Cloud Pub/Sub to perform some real-time calculations.”

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"We are currently using Airflow workflow management on Compute Engine as our hosted ETL platform, but are likely to move to Cloud Composer in future."

500 million records per day

With Google Cloud providing its data infrastructure, ViSenze is well positioned to meet internal and customer demands for more granular insights. The business is now processing 500 million records per day through BigQuery and saves up to one year's aggregated data—excluding any personal data—in the data warehouse for analysis. The nature of ViSenze's business means most reports are generated for data processed on an hourly, daily, or monthly basis. "BigQuery is extremely stable and

performance optimized, regardless of the volume of data it processes,” says Yao. “Across BigQuery and other Google Cloud Platform services, we’ve recorded 99.99% availability over the past year.”

The lack of complexity and the ease of use of BigQuery has enabled ViSenze to reduce its data infrastructure and management costs by 30%–50%, and scale up without incurring downtime. “With BigQuery, we have saved the equivalent of two full-time engineers and now need only half of one person’s time to maintain our whole data platform,” says Yao. “In addition, BigQuery integrates closely with Looker Studio, enabling non-technical people in our product and business teams to create dynamic, detailed analysis dashboards. We now use Looker Studio to create nearly 50 separate reports.”

The business has now grown to offer access to more than 1 billion users and a listing of more than 400 million purchasable products.

Next steps

ViSenze is now researching the potential of the Cloud AutoML (<https://cloud.google.com/automl/>) suite of machine learning products to improve the training of its models and run a fully managed NoSQL database through Cloud Datastore (<https://cloud.google.com/datastore/>). “A NoSQL database service is the only missing piece of our architecture for now, and using Cloud Datastore would enable us to focus almost exclusively on our business,” says Yao. “With Google Cloud Platform, we are ideally positioned to continue providing

support to our business team and help them continue expanding into new markets.

“In addition, we can help retailers and consumers to unlock the potential of the web and apps to transform the purchasing experience.”